

Supplementary Material: Full-Cohort Revision

Validation, missingness, node splits, and model status

Raw Data Verification

{ "data_root": "/Users/moe/Programming/Thesis-curser/mimic-iv-3.1", "manifest_exists": true, "manifest_entries": 33, "files_present": 33, "files_missing": [], "required_files_present": { "hosp/patients.csv.gz": true, "hosp/admissions.csv.gz": true, "hosp/diagnoses_icd.csv.gz": true, "hosp/labevents.csv.gz": true, "icu/icustays.csv.gz": true, "icu/chartevents.csv.gz": true }, "complete_by_manifest": true }

Model Completion Status

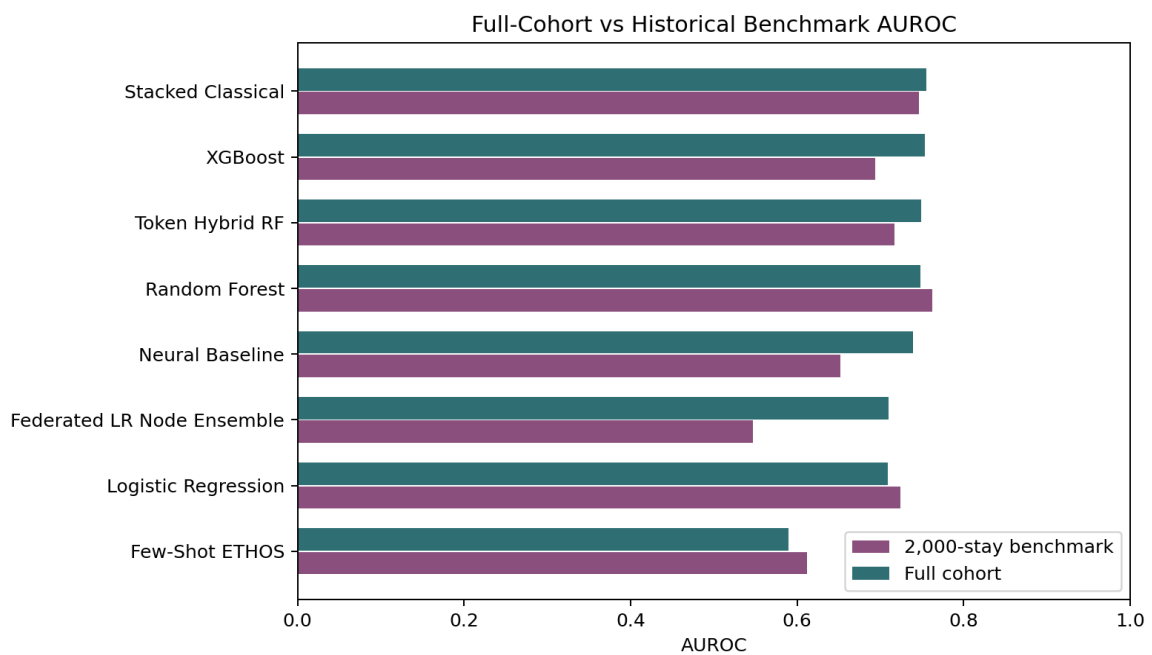
Model/component	Status	Details
Logistic Regression	completed	seconds: 1.692
Random Forest	completed	seconds: 5.367
Neural Baseline	completed	seconds: 3.188
XGBoost	completed	seconds: 2.421
LightGBM	completed	seconds: 5.891
CatBoost	completed	seconds: 3.744
Stacked Classical	completed	seconds: 12.394
Token Hybrid RF	completed	seconds: 4.165
Federated LR Node Ensemble	completed	seconds: 1.629; nodes_used: 5
Few-Shot ETHOS	completed	seconds: 10.586; episodes: 40

Full-Cohort Model Metrics

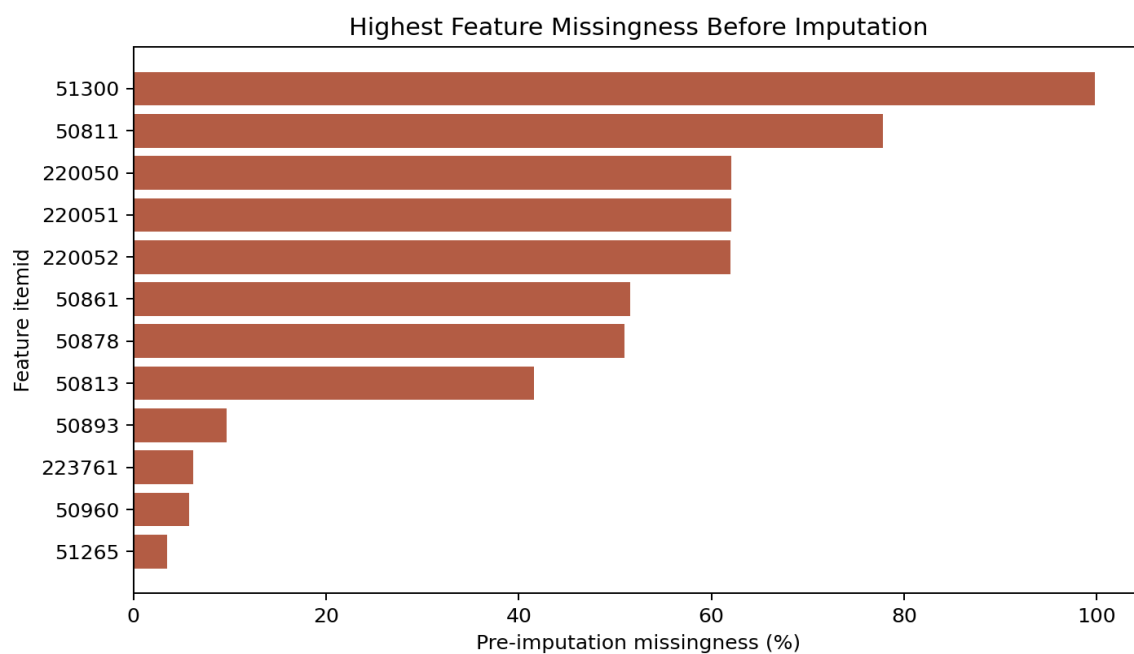
Model	Train/Val/Test	AUROC	AUPRC	Accuracy	Recall	Specificity	F1	Brier	ECE
Stacked Classical	20735/5184/6480	0.755	0.825	0.699	0.769	0.583	0.677	0.202	0.109
XGBoost	20735/5184/6480	0.753	0.825	0.697	0.761	0.589	0.675	0.189	0.020
CatBoost	20735/5184/6480	0.753	0.823	0.703	0.790	0.558	0.677	0.200	0.100
LightGBM	20735/5184/6480	0.751	0.823	0.690	0.730	0.623	0.674	0.199	0.090
Token Hybrid RF	20735/5184/6480	0.749	0.820	0.698	0.772	0.573	0.674	0.192	0.031
Random Forest	20735/5184/6480	0.748	0.820	0.689	0.742	0.601	0.670	0.192	0.028
Neural Baseline	20735/5184/6480	0.739	0.814	0.675	0.682	0.663	0.664	0.194	0.015
Federated LR Node Ensemble	20735/5184/6480	0.709	0.778	0.678	0.746	0.565	0.656	0.215	0.109
Logistic Regression	20735/5184/6480	0.709	0.778	0.680	0.754	0.556	0.656	0.215	0.109
Few-Shot ETHOS	20735/5184/6480	0.590	0.668	0.590	0.642	0.504	0.570	0.250	0.127

Historical Benchmark Comparison

Model	Full AUROC	Benchmark AUROC	Delta	Benchmark source
Stacked Classical	0.755	0.746	0.009	2,000-stay stacked BEST
XGBoost	0.753	0.694	0.059	2,000-stay Exp1
CatBoost	0.753	NA	NA	No directly comparable old benchmark metric
LightGBM	0.751	NA	NA	No directly comparable old benchmark metric
Token Hybrid RF	0.749	0.717	0.032	2,000-stay ETHOS-RF hybrid
Random Forest	0.748	0.762	-0.015	2,000-stay Exp1
Neural Baseline	0.739	0.652	0.087	2,000-stay Exp1
Federated LR Node Ensemble	0.709	0.547	0.162	2,000-stay federated experiment
Logistic Regression	0.709	0.724	-0.015	2,000-stay Exp1
Few-Shot ETHOS	0.590	0.611	-0.022	2,000-stay Exp1



Missingness



Federated Node Distribution

node	rows	unique_stays	feasible_0	feasible_1
node_1	8333	8333	3752	4581
node_2	7432	7432	2530	4902
node_3	5826	5826	2096	3730
node_4	5268	5268	1849	3419
node_5	5540	5540	1875	3665